MCS Saturday SAT Math Homework

Instructor: Hsiao 'Hue' Weng · weng.hue.8@gmail.com

Linear Expressions + Equations

Digital SAT Math · Session 3

Distribute Fully · Isolate Precisely · Homework Activity Sheet

Name _____ **Date** _____

Completion Check Score after review; justify every correction in the error log.	____ / 18 Core	____ / 6 Bridge	____ / 24 Total

A Structure Before Symbols

4 prompts · 7 min

Do not solve unless asked. Name the legal move and explain what it preserves.

A1 · Distribution — Explain

For **−4(2x + 7)**, describe exactly how the factor **−4** changes **both** terms. Why must the constant change sign?

State the two products and the sign logic.

(−4)(2x) = −8x
(−4)(+7) = −28
Answer: −8x − 28.
Multiply both terms by −4; the constant becomes negative because negative × positive = negative.

A2 · Like Terms — Classify

From **5x², −3x, 8, 11x², 2x**, list the groups that may legally combine. Explain why **x²**-terms and **x**-terms are not alike.

Group first; then defend the boundary.

Groups: {5x², 11x²}, {−3x, 2x}, {8}.
Combined: 16x² − x + 8.
Like terms have the same variable part and exponent; x² and x have different exponents.

A3 · Equality — Audit

A student changes **6x − 11 = 31** to **6x = 31**. Identify the first invalid step and write the correct next line.

Name what must happen to both sides.

Invalid: 11 was added only to the left.
Add 11 to both sides:
6x − 11 + 11 = 31 + 11
Correct next line: 6x = 42.

A4 · Invariant — Explain

In one precise sentence, distinguish **rewriting one side equivalently** from **performing an operation on both sides**.

Use the phrase “solution set.”

An equivalent rewrite keeps one side’s value unchanged, while a reversible operation applied to both sides changes their forms together; both preserve the solution set.

B Translate and Defend

4 prompts · 9 min

Define the variable with units, write an equation, solve, and interpret the result.

B1 · Context Model — Construct

A bicycle rental costs a one-time safety fee of **$9** plus the same hourly rate. A 6-hour rental costs **$57**. Let **r** be the hourly rate in dollars per hour. Write and solve an equation for **r**.

Define → model → solve → check units.

Let r be the hourly rate in dollars per hour.
9 + 6r = 57
6r = 48
r = 8 dollars per hour.
Check: 9 + 6(8) = 57 dollars; 6 hours × $8/hour = $48.

Hourly rate: **$8 per hour**

B2 · Meaning — Explain

In your equation from B1, explain which amount occurs once, which amount repeats, and why the coefficient is 6.

The $9 safety fee occurs once.
The charge r dollars repeats each hour.
The coefficient 6 is the number of hours, so 6r is the total hourly charge.

B3 · Representation — Choose

Which equation represents: “Seven less than three times a number is 20”?

A. $7 - 3x = 20$
B. $3(x - 7) = 20$
C. $3x - 7 = 20$
D. $7x - 3 = 20$

C. $3x - 7 = 20$
“Three times a number” is 3x; “seven less than” that quantity means subtract 7 from 3x.

B4 · Distractor Autopsy — Analyze

For B3, choose one incorrect option and describe the exact language error that would produce it.

Do not merely say “wrong order.” Connect words to symbols.

Example: A reverses the subtraction. “Seven less than 3x” means start with 3x and subtract 7, giving $3x - 7$. The expression $7 - 3x$ instead means “three times a number subtracted from seven.”

C Linear Fluency Lab

8 questions · 14 min

Write one legal move per line. Substitute every answer into the original equation before moving on.

Distribute completely · Combine within each side · Create one variable side · Plug back

C1 Level 1

8x + 13 = 61

8x = 48
x = 6
Check: 8(6) + 13 = 61.

x = **6**

C2 Level 1

4(3x − 2) = 52

3x − 2 = 13
3x = 15
x = 5
Check: 4(3·5 − 2) = 52.

x = **5**

C3 Level 2

15 − 2(4x + 1) = −19

−2(4x + 1) = −34
4x + 1 = 17
4x = 16
x = 4
Check: 15 − 2(4·4 + 1) = −19.

x = **4**

C4 Level 2

7x + 18 = 3x − 10

4x + 18 = −10
4x = −28
x = −7
Check: 7(−7) + 18 = −31;
3(−7) − 10 = −31.

x = **−7**

C5 Level 2

5(2x + 3) − 4 = 3(x + 9)

10x + 15 − 4 = 3x + 27
10x + 11 = 3x + 27
7x + 11 = 27
7x = 16
x = 16/7
Check: both original sides equal 237/7.

x = **16/7**

C6 Level 3

4 − 3(2x − 5) = 5(x + 1) + 3

4 − 6x + 15 = 5x + 5 + 3
19 − 6x = 5x + 8
19 = 11x + 8
11 = 11x
x = 1
Check: 4 − 3(2 − 5) = 5(1 + 1) + 3 = 13.

x = **1**

C Linear Fluency Lab (continued)

8 questions · 14 min

Write one legal move per line. Substitute every answer into the original equation before moving on.

Distribute completely · Combine within each side · Create one variable side · Plug back

C7 Classify

Determine whether the equation has **one solution, no solution, or infinitely many solutions**.

$6(x - 2) + 4 = 2(3x - 4)$

$6x - 12 + 4 = 6x - 8$
$6x - 8 = 6x - 8$
$-8 = -8$
Both sides agree for every real x.

Classification: **Infinitely many solutions**

C8 Classify

Determine whether the equation has **one solution, no solution, or infinitely many solutions**.

$9 - 4(2x + 1) = 5 - 8x$

$9 - 8x - 4 = 5 - 8x$
$5 - 8x = 5 - 8x$
$5 = 5$
Both sides agree for every real x.

Classification: **Infinitely many solutions**

D Equivalent or Not?

3 audits · 6 min

For each proposed step, mark Valid or Invalid. If invalid, repair only that step.

D1 · Distribution Audit

$-3(5x - 4) \rightarrow -15x - 12$

A. Valid **B.** Invalid

Repair or justify.

Invalid. Repair: $-15x + 12$.
$(-3)(-4) = +12$; a negative times a negative is positive.

D2 · Both Sides Audit

$2x + 9 = 17 \rightarrow 2x = 8$

A. Valid **B.** Invalid

Name the operation.

Valid. Subtract 9 from both sides:
$2x + 9 - 9 = 17 - 9$
$2x = 8$.

D3 · Combining Audit

$4x^2 + 7x - 3x^2 \rightarrow 8x^2$

A. Valid **B.** Invalid

Rewrite the expression correctly and explain the variable parts.

Invalid. Repair: $x^2 + 7x$.
Combine only $4x^2$ and $-3x^2$. The term $7x$ has exponent 1, so it is not like an x^2-term.

E Guardrail Bridge

6 questions · 12 min

Use the shortcut only when its condition is satisfied. For radicals, generate candidates and then return to the original equation.

E1 · Proportion — Solve

$(x + 5)/6 = (3x - 1)/10$

Multiply both sides by 30:
$5(x + 5) = 3(3x - 1)$
$5x + 25 = 9x - 3$
$25 = 4x - 3$
$28 = 4x$
$x = 7$
Check: $12/6 = 20/10 = 2$.

$x =$ **7**

E2 · Permission — Explain

A student wants to "cross-multiply" in **$(x + 2)/3 + 4 = 7$**. Explain why the diagonal shortcut does not apply as written, then give a legal first move.

The left side is a sum, not one fraction as written; the +4 cannot be ignored.
Legal first move: subtract 4 from both sides.
$(x + 2)/3 = 3$

E3 · Radical — Solve + Check

$\sqrt{(3x + 4)} = x$

Conditions: $3x + 4 \geq 0$ and $x \geq 0$.
$3x + 4 = x^2$
$x^2 - 3x - 4 = 0$
$(x - 4)(x + 1) = 0$
Candidates: 4 and -1.
Check 4: $\sqrt{16} = 4$. Accept.
Check -1: $\sqrt{1} = 1 \neq -1$. Reject.

Valid solution(s): **4 only**

E4 · Domain — Reason

Before squaring **$\sqrt{(5x - 6)} = x - 4$**, what two conditions must be true? State each as an inequality.

Check the radicand and the right side.

$5x - 6 \geq 0$ (real radicand).
$x - 4 \geq 0$ (nonnegative right side).
Equivalently, $x \geq 6/5$ and $x \geq 4$.
Together: $x \geq 4$.

E5 · Candidate Trial — Verify

Algebra produces candidates **$x = 2$** and **$x = 7$** for **$\sqrt{(x + 2)} = x - 4$**. Test both in the **original** equation and report the verdict.

$x = 2$: $\sqrt{(2 + 2)} = 2$, but $2 - 4 = -2$.
$2 \neq -2$: reject 2.
$x = 7$: $\sqrt{(7 + 2)} = 3$ and $7 - 4 = 3$.
$3 = 3$: accept 7.
Valid solution: $x = 7$.

E6 · Error Mechanism — Explain

Why can squaring both sides create a solution that was not valid before squaring? Use sign information in your explanation.

Squaring erases sign information: opposite numbers have the same square. For example, $2^2 = (-2)^2$ even though $2 \neq -2$. A squared equation may therefore admit an extraneous candidate; test in the original equation.

F Digital SAT Timed Set

6 questions · 9 min

Set a 9-minute timer. Answer every item. For graph verification, identify the requested coordinate before opening Desmos.

Box the deliverable · Choose algebra or graph · Audit signs · Zero blanks

F1 Level 1

Which expression is equivalent to **3(4x − 5) + 2x**?

A. 14x − 5

B. 14x − 15

C. 12x − 13

D. 6x − 15

B. 14x − 15
3(4x − 5) + 2x = 12x − 15 + 2x
= 14x − 15.

F2 Level 2

If **11 − 3(2x − 1) = −22**, what is the value of **x**?

−3(2x − 1) = −33
2x − 1 = 11
2x = 12
x = 6
Check: 11 − 3(12 − 1) = −22.

SPR: **6**

F3 Level 2

A meal-delivery plan charges a one-time cooler deposit of **$24** and **$11** per meal. A customer's first order costs **$101**. How many meals were ordered?

Let n be the number of meals.
24 + 11n = 101
11n = 77; n = 7 meals.
Check: 24 + 11(7) = 101.

SPR: **7**

F4 Level 3

For what value of **k** does **k(2x − 3) = 10x − 15** have infinitely many solutions?

A. 2

B. 3

C. 5

D. 15

C. k = 5
10x − 15 = 5(2x − 3).
With k = 5, both sides are identical for every real x.

F5 Level 3

The lines **y = 4x − 9** and **y = −2x + 15** intersect at **(a, b)**. What is the value of **a**?

Algebra or Desmos verification.

4x − 9 = −2x + 15
6x − 9 = 15
6x = 24; x = 4, so a = 4.
Check: both lines give y = 7.
Intersection: (4, 7); read x, not y.

SPR: **4**

F6 Level 4

If **(m + 2)x + 7 = 9x + 7** is true for every value of **x**, what is **m**?

Match coefficients: m + 2 = 9.
m = 7.
Check: substituting m = 7 gives
9x + 7 = 9x + 7 for every x.

SPR: **7**

G Verification Record

Evidence before answers

Complete this section before checking an answer key or discussing solutions.

- ☐ I distributed to every term, including constants.
- ☐ I kept equations equivalent one line at a time.
- ☐ I stated conditions before using an advanced shortcut.
- ☐ I answered the requested quantity, coordinate, or unit.
- ☐ I combined only terms with identical variable parts.
- ☐ I checked each linear solution in the original equation.
- ☐ I tested every radical candidate in the original equation.
- ☐ I finished the timed set with zero blanks.

Timing Evidence Record

How long did F1–F6 take? Which question consumed the most time, and what observable trigger should have made you flag it?

Student-specific response frame:
I took ___ minutes ___ seconds.
F___ took the longest. I should have flagged it when ___.
Example trigger: repeating algebra without progress.

Tool Evidence Record

If you used Desmos, state the two expressions entered, the graph feature inspected, and the exact value read. If not, identify one item it could verify.

Model: F5 could be verified by graphing $y = 4x - 9$ and $y = -2x + 15$. Their intersection is (4, 7); read the first coordinate, a = 4.

H Error Log

Up to 5 corrections

Choose one primary error type: Distribution, Like Terms, Equality, Translation, Condition, Checking, Reading, or Timing.

Illustrative entries only — use rows that match the student's actual errors.

#	Problem	My original thought	Primary error type	Correction rule
1	C6	$-3(-5) = -15$.	Distribution	Write each signed product; negative × negative = positive.
2	D3	$4 + 7 - 3 = 8$, so $8x^2$.	Like Terms	Group by variable and exponent before combining coefficients.
3	B3	"Seven less than 3x" means $7 - 3x$.	Translation	Start with 3x, then subtract 7.
4	E5	Both roots of the squared equation are answers.	Checking	Substitute every candidate into the original unsquared equation.
5	F5	The intersection has y = 7, so enter 7.	Reading	Name the requested coordinate first: a is x.

I Transfer Reflection

Complete sentences

1. The first visible sign that I have made a distribution or sign error is…

The first visible sign that I have made a distribution or sign error is that a term inside the parentheses has not been multiplied by the outside factor, or that I have lost the sign of a product; in C6, the product $-3(-5)$ must be $+15$.

2. One equation in this homework that I can now solve by two representations is… The representations are…

One equation in this homework that I can now solve by two representations is $11 - 3(2x - 1) = -22$ in F2. The representations are an algebraic equation solved by inverse operations and two graphs whose intersection is $(6, -22)$; both give $x = 6$.
For the graph, enter $y = 11 - 3(2x - 1)$ and $y = -22$, then read the intersection's x-coordinate.

3. The distinction between a candidate and a verified solution matters because…

The distinction between a candidate and a verified solution matters because a transformation such as squaring can introduce a value that does not satisfy the original equation; in E5, $x = 2$ passes the squared equation but gives $2 \neq -2$ in the original.

4. On my next Digital SAT Math module, the correction rule I will apply first is…

On my next Digital SAT Math module, the correction rule I will apply first is to box the requested quantity before choosing a method, so I do not report a y-coordinate when the question asks for an x-coordinate, as in F5.